

$\binom{n}{k} = \frac{n!}{k!(n-k)!}$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

ex a die is rolled to yield "i", then a coin is tossed the same # of times. $P(\text{head not appear}) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + \dots$

B_k : get "i" for the k-th roll. $P(A|B_k) = (\frac{1}{2})^k$

Bernoulli(p) $P(k) = p^k(1-p)^{1-k}$ $EX = p$ $VX = p(1-p)$

Binomial(p) $P(k) = \binom{n}{k} p^k (1-p)^{n-k}$ $EX = np$ $VX = np(1-p)$

Geometric(p) $P(k) = (1-p)^{k-1} p$ $EX = \frac{1}{p}$ $VX = \frac{1-p}{p^2}$

NB(r, p) $P(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$ $EX = \frac{r}{p}$ $VX = \frac{r(1-p)}{p^2}$

Pois(λ) $P(k) = \frac{\lambda^k e^{-\lambda}}{k!}$ $EX = \lambda$ $VX = \lambda$

Uni(a, b) $f(x) = \frac{1}{b-a}$ $F(x) = \frac{x-a}{b-a}$ $EX = \frac{a+b}{2}$ $VX = \frac{(b-a)^2}{12}$

exp(λ) $f(x) = \lambda e^{-\lambda x}$ $F(x) = 1 - e^{-\lambda x}$ $EX = \frac{1}{\lambda}$ $VX = \frac{1}{\lambda^2}$

Normal(μ, σ^2) $f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$

$cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = EXY - EX\mu_Y - \mu_X EY$

$Var(X+Y) = Var(X) + Var(Y) + 2cov(X, Y)$

$cov(X, Y) = \frac{cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$

$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$

$F_{X|Y}(x|y) = \int_{-\infty}^x f(u, y) du / f_Y(y)$

$Var(Y|X) = E[(Y - E(Y|X))^2 | X]$

$Var(Y) = Var(E(Y|X)) + E(Var(Y|X))$

ex. X general discrete $P(X=x_k) = P_k$

define $F_k = P_1 + \dots + P_k$ $k=1, 2, \dots, n$

output $X = x_k$ if U lies in $(F_{k-1}, F_k]$

- $U \sim U(0,1)$; $F_0 = 0$

- $k=0$

- while $U > F_k$

- output $X = x_k$

ex Generate Binomial(n, p)

① $U \sim U(0,1)$, $F_1 = 0$ ② use the interpretation

- $k=1$

- while $U > F_k$

- output $X = k$

$P(X \leq m) = 1 - (1-p)^m$ Geom(p) use interpretation

② JTM $F_k = 1 - P(X > k) = 1 - (1-p)^k$

$X = \min \{k: U \leq F_k\} = \lceil \frac{\log(1-U)}{\log(1-p)} \rceil$

if p small, ① can be slow because it takes many iterations to get 1st success

if p big, ① should be fast as it only has small # of iterations

ex Poi(λ) $P(k+1) = \frac{\lambda}{k+1} P(k)$

$V = \text{rand}(0,1)$ $F(k+1) = F(k) + P(k+1)$

$P = \exp(-\lambda)$ $F(0) = P(0)$ $F(1) = P(0) + P(1)$

$v=0$ while $V > F$

$P = \lambda P / (1+P)$ $F(1+1) = F(1) + P(1+1)$

$F = F + P$ $i=i+1$

output i

Binom(n, p) $P(k+1) = \frac{n-k}{k+1} \cdot \frac{p}{1-p} \cdot P(k)$

NB(r, p) ① $P(k+1) = \frac{k(1-p)}{k+1-r} P(k)$

② for $i=1, \dots, r$ ③ interpretation

$Y \sim \text{Geom}(p)$ $n=0$ $S=0$

$X = X+Y$ while $S < r$

output X $B \sim \text{Bern}(p)$ $S = S+B$

$n = n+1$ return n

continuous RV $\rightarrow$ JTM $\checkmark$

ex $f(x) = \begin{cases} x, & x \in [0,1] \\ 2-x, & x \in [1,2] \\ 0, & \text{else} \end{cases}$ $P(x) = \begin{cases} \frac{x^2}{2}, & x \in [0,1] \\ -\frac{(2-x)^2}{2} + 1, & x \in [1,2] \\ 1, & x \geq 2 \end{cases}$

$\rightarrow x \in [0,1]$ $F(x) \in [0, \frac{1}{2}] \rightarrow u = \frac{x^2}{2}$ for $0 \leq u \leq \frac{1}{2}$

$x \in [1,2]$ $F(x) \in [\frac{1}{2}, 1] \rightarrow u = -\frac{(2-x)^2}{2} + 1$ for $\frac{1}{2} \leq u < 1$

$X \sim U(a, b)$ $X = a + (b-a)U$

写清楚 support

泊松分布 - 一种分布 $(p = u_1 u_2 \dots u_k \geq e)$

1. $L = e^{-\lambda}$, $p=1, k=0$ 取对数等价为

2. $u \sim \text{Uni}$ 指数分布累加

3. $p = p \cdot u$

p. if $p > L$, $k=k+1$, go back to 2

else output $X = k$

$\int_0^\infty e^{-t^2} dt = \sqrt{\frac{\pi}{2}}$

A-R method Find a constant c such that $\frac{f(x)}{g(x)} \leq c$ for all x of c > 1 有关的

Algorithm 1. Generate $Y \sim g(y)$ $\Rightarrow g(x)$ support 覆盖 fix

2. Generate $V \sim \text{Uni}(0,1)$

3. if $V \leq \frac{f(Y)}{cg(Y)}$, set $X=Y$ (accept)

else go back to the beginning (reject)

4. output X $\Rightarrow g(x) \sim T \text{ pdf}$

ex $f(x) = x^2$ $0 \leq x \leq 1$ let $g(x) = 1$ $c=3$

$U(0,1)$ sequence $0.1, 0.32, 0.5, 0.49, 0.23$

$Y=0.1$ $V=0.32 > Y^2$ reject

$Y=0.5$ $V=0.49 > Y^2$ reject

ex $N(\mu, \sigma^2)$ let $g(x) = \frac{1}{\sigma} e^{-|x|}$

$\frac{f(x)}{g(x)} = \sqrt{\frac{2}{\pi}} e^{-\frac{x^2}{2} + |x|} \leq c$ $c = \sqrt{\frac{2}{\pi}} e^{\frac{1}{2}}$

$g(x)$? generate $V = \begin{cases} +1 & \text{w.p. } \frac{1}{2} \\ -1 & \text{w.p. } \frac{1}{2} \end{cases}$

$W \sim \exp(1)$ $\rightarrow V \cdot W$

$p(\text{accept}) = \int_{-\infty}^{\infty} P(\frac{f(x)}{cg(x)} > u) g(x) dx = \frac{1}{c}$

of iterations $\sim \text{geom}(\frac{1}{c})$

mean # of iterations = c (c close to 1 is good)

$f(x) = \frac{1}{k} h(x)$ $\frac{h(x)}{g(x)} = \tilde{c}$

can estimate k by: $E(\text{runtime}) = c = \frac{1}{p}$ $\hat{k} = \frac{\tilde{c}}{c} = \tilde{c} \hat{p}$

ex $f(x) = \frac{1}{k} e^{-\frac{x^2}{2}} e^{\sin^2 x}$ $x \in \mathbb{R}$

let $g(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ $\frac{h(x)}{g(x)} \leq \sqrt{2\pi} e$

ex gamma(n,1) $f(x) = \frac{e^{-x}}{\Gamma(n)} \frac{x^{n-1}}{(n-1)!}$ $x \geq 0$

$g(x) = \frac{1}{2} e^{-\frac{x}{2}}$ (JTM)

$\frac{f(x)}{g(x)} = 2 e^{-\frac{x}{2}} \frac{x^{n-1}}{(n-1)!}$ let $h(x) = e^{-\frac{x}{2}} x^{n-1}$

$\log h(x) = -\frac{x}{2} + (n-1) \log x$ $\frac{d \log h(x)}{dx} = 0 \Rightarrow x = 2(n-1)$

$c = \left(\frac{f(x)}{g(x)} \right)_{\max} = 2 \cdot \frac{1}{(n-1)!} \cdot h(2(n-1))$

ex $f(x) = \frac{2}{\pi} \sqrt{1-x^2}$ $x \in [-1,1]$

let $g(x) = \frac{1}{2} (x^2 + 1 - x)$ $\frac{f(x)}{g(x)} \leq \frac{4}{\pi}$

$x \in [-1,1]$ $U \sim \text{Uni}(0,1) \rightarrow 2V-1$

$\frac{f}{cg} \approx 1$ high prob accept Y

Uniform Sampling on Irregular Regions $A = \{(y, w): a \leq y \leq b, 0 \leq w \leq h(y)\}$

① K bound $x \sim \text{Uni}(a,b) \rightarrow b-a$

$w \sim \text{Uni}(0, h(y))$

output (Y, w)

Now: sample using *

accept if $h(y) \leq \tilde{K}$ (fail in A)

② sample from $\frac{h(y)}{K}$ $a \leq y \leq b$

where K is a (possibly unknown) constant

given Y, sample w from $\text{Uni}(0, h(Y))$

"faster because no need to reject"

why work? $f_{Y,W}(y, w) = \begin{cases} \frac{1}{K}, & a \leq y \leq b, 0 \leq w \leq h(y) \\ 0, & \text{else} \end{cases}$

where K is the area of A

1. marginal pdf of Y $f_Y(y) = \int_0^{h(y)} \frac{1}{K} dw = \frac{h(y)}{K}$

2. conditional on w/y $f_{W|Y}(w|y) = \frac{f_{Y,W}(y, w)}{f_Y(y)} = \frac{1}{h(y)}$

ex $A = \{(y, w): 0 \leq y \leq 1, 0 \leq w \leq \cos y + 1\}$

$\Rightarrow K = \int_0^1 \int_0^{\cos y + 1} dw dy = \int_0^1 (\cos y + 1) dy = \frac{1}{2}$

generate $Y \sim h(Y)$ $\frac{\cos y + 1}{K} = \frac{\cos y + 1}{\frac{1}{2}}$

given Y=y, generate $w \sim U(0, h(y))$

A-R let $g(y) = \frac{1}{4y^2}$ for $0 \leq y \leq 1$

JTM ① $U \sim \text{Uni}(0,1)$

② $\hat{y} = U^{\frac{1}{2}}$

③ $V \sim \text{Uni}(0,1)$ $\frac{\cos y + 1}{y^2}$

④ if $V < \frac{h(y)}{cg(y)} = \frac{\cos y + 1}{\frac{1}{4y^2}} = 4y^2(\cos y + 1)$

set $y = \hat{y}$, otherwise go back to ④

⑤ $X \sim \text{Uni}(0,1)$

⑥ $W = x h(y) = x \cdot \frac{\cos y + 1}{y^2}$

ex $f(x) = \begin{cases} \frac{1}{\pi} \sqrt{1-x^2}, & x \in [-1,1] \\ 0, & \text{else} \end{cases}$

M4 Q1 Let G be a c.d.f

$F(x) = \frac{G(x) - G(a)}{G(b) - G(a)}$ $a \leq x \leq b$

(a) show $P(Y \leq x | X \leq b) = \frac{G(x) - G(a)}{G(b) - G(a)}$

Composition Method

discrete case

$$P(X=x) = \sum_{j=1}^m P(Y_j \leq x) P(J=j)$$

To generate X:

- generate J
- given J=j, generate Yj

ex $P(X=x) = \begin{cases} 0.05, & x=1,3,\dots,5 \\ 0.15, & x=6,7,\dots,10 \end{cases}$

Define J as

$$P(J=1) = 0.05 \times 5 = 0.25$$

$$P(J=2) = 0.15 \times 5 = 0.75$$

Define

$$P(Y_j=1) = \dots = P(Y_j=5) = \frac{1}{5}, j=1 \text{ else } X = \lfloor 5V \rfloor + 6$$

$$P(Y_j=6) = \dots = P(Y_j=10) = \frac{1}{5}, j=2$$

continuous case

$$P(X=x) = \int_0^\infty P(Y_t \leq x) f_T(t) dt$$

$$f(x) = \int_0^\infty f_{Y_t}(x) f_T(t) dt$$

X is a mixture of $Y_t: t \geq 0$

ex $f(x) = \frac{1}{2} e^{-|x|} - \infty < x < \infty$

$$J=1 \text{ u.p. } \frac{1}{2} \quad Y_j \sim \exp(1)$$

$$J=2 \text{ u.p. } \frac{1}{2} \quad Y_j \sim -\exp(1)$$

Algo:

- generate J from $(\frac{1}{2}, \frac{1}{2})$ on $\{1, 2\}$
- generate $Y \sim \exp(1) \rightarrow \text{ITM}$
- output J.Y

check:

$$\text{for } x > 0 \quad f(x) = P(J=1) \cdot f_{Y_1}(x) = \frac{1}{2} e^{-x} \quad Y_1 \sim \exp(1)$$

$$\text{for } x < 0 \quad f(x) = P(J=2) \cdot f_{Y_2}(x) = \frac{1}{2} e^x \quad Y_2 \sim -\exp(1)$$

$$\textcircled{c} \text{ ITM. } F(x) = \begin{cases} \frac{1}{2} e^x & x < 0 \\ 1 - \frac{1}{2} e^{-x} & x \geq 0 \end{cases}$$

$$X = \text{sgn}(U-0.5) \ln(1-2|U-0.5|) \text{ else } \dots$$

$$\textcircled{d} X = X_1 - X_2 \quad X_1 \sim \exp(1) \quad X_2 \sim \exp(1)$$

ex $f(x) = \int_0^\infty 2e^{-\frac{x}{2}-t^2} dt, x > 0; t > 0$

normalize $f_{Y_t}(x)$!

$$\text{Let } f_{Y_t}(x) = \frac{1}{t} e^{-\frac{x}{t}} \text{ for } x > 0 \quad Y_t \sim \exp(\frac{1}{t})$$

$$\text{then } f_T(t) = 2te^{-t^2} \text{ for } t > 0$$

$$\text{Generate } T \sim f_T(t) \quad X = T \cdot \log(U)$$

$$\text{- Given } T=t, \text{ generate } Y_t \sim \exp(\frac{1}{t})$$

$$\text{alt } F_T(t) = \int_0^t 2s e^{-s^2} ds = 1 - e^{-t^2} = U$$

ex $P(X=j) = (\frac{1}{2})^{j+1} + (\frac{1}{2}) (\frac{2^{j-1}}{3^j}) \quad j=1,2,\dots$

$$\textcircled{1} - V \sim \text{unif}(0,1)$$

$$- V \sim \text{unif}(0,1)$$

$$\text{if } U \leq 0.5 \quad X = \left\lceil \frac{\log(1-V)}{\log(1-\frac{1}{3})} \right\rceil$$

$$\text{else } X = \left\lceil \frac{\log(1-V)}{\log(1-\frac{1}{3})} \right\rceil$$

ITM

$$F(x) = P(X \leq x) = \frac{1}{2} (1 - (\frac{1}{3})^x) + \frac{1}{2} (1 - (\frac{2}{3})^x)$$

$$\text{- generate } U \sim \text{unif}(0,1)$$

$$\text{- Find the smallest integer } x \geq 1 \text{ such that } F(x) \geq U$$

AR for $(X, Y) \quad \forall x, y \in \mathbb{R}$

ex $f(x, y) = \frac{1}{\pi} \exp(-x^2 - y^2 + x \sin(\pi y))$

$$\textcircled{1} \text{ Let } g(x, y) = (\frac{1}{2} e^{-|x|}) (\frac{1}{2} e^{-|y|})$$

$$\frac{f(x, y)}{g(x, y)} = 4 \exp(-(x-1)^2 - (y-1)^2) \leq 4e^2$$

$$\textcircled{2} \text{ Let } g(x, y) = C \exp(-\frac{1}{2} x^2) \exp(-y^2)$$

$$x \sim N(0,1), y \sim N(0, \frac{1}{2})$$

$$\frac{h(x, y)}{g(x, y)} = \frac{1}{2} \exp(-\frac{1}{2} x^2 + x \sin(\pi y)) \leq \exp(\frac{1}{2}) \leq C$$

$$\text{AR for } (X, Y) \quad p(x=i) = \text{pois}(x) \frac{e^{-x} x^i / i!}{\sum_{j=0}^{\infty} e^{-x} x^j / j!}$$

$$i=0,1,2,\dots,k \quad \frac{e^{-x} x^i / i!}{\sum_{j=0}^{\infty} e^{-x} x^j / j!}$$

$$\text{To find } C \cdot \frac{f(x)}{g(x)} \leq C \text{ for } i=0,\dots,k$$

$$\text{Let } C = (\sum_{j=0}^{\infty} \frac{e^{-x} x^j}{j!})^{-1}$$

$$\text{- generate } Y \sim \text{Pois}(x)$$

$$\text{- "accept } Y" \text{ u.p. } 1 \text{ for } y=0,\dots,k$$

$$\text{and } \frac{f(y)}{g(y)} = 0 \text{ for } y \text{ not in } \{0,\dots,k\}$$

$$\text{if } y \text{ is not in } \{0,\dots,k\} \text{ then } \frac{f(y)}{g(y)} = 0$$

$$\text{if } y \text{ is in } \{0,\dots,k\} \text{ then } \frac{f(y)}{g(y)} = \frac{e^{-x} x^y / y!}{\sum_{j=0}^{\infty} e^{-x} x^j / j!}$$

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Uniform Sampling

ex $A = \{(x, y) : x \in [1,1], |y| \leq x\}$

$$\textcircled{1} X \textcircled{2} V \quad (\text{if } V \leq x \text{ then } Y = xV \text{ else } Y = -xV)$$

$$\text{ex } A = \{x \in [0,1], 0 \leq y \leq x^{-1}\} \quad h(x) = x^{-1}$$

$$f(x) = \frac{h(x)}{K} = \frac{h(x)}{\int h(x) dx} = \frac{1}{2} x^{-1}$$

$$F(x) = x^{\frac{1}{2}} \quad x \in [0,1] \rightarrow \text{ITM}$$

$$\text{then } Y \sim \text{unif}(0, h(x))$$

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$$\text{ex } g(x) = \frac{1}{x^2} \quad 0 \leq x \leq 1$$

$$\text{ex } g(x) = \frac{1}{x^2} \quad x \geq 1$$

$$A = \int_0^1 g_1(x) dx + \int_1^\infty g_2(x) dx = A_1 + A_2$$

$$= \frac{1}{2} + \frac{1-a}{a} = \frac{2-a}{a} \quad \text{Fit uniform } K=A$$

$$\text{w.p. } \frac{A_1}{A} \quad X \sim \frac{g_1(x)}{A_1} \quad X \sim \frac{1}{2-a} x^{\frac{1}{2-a}} (0,1)$$

$$\text{w.p. } \frac{A_2}{A} \quad X \sim \frac{g_2(x)}{A_2} \quad X \sim \frac{1}{2-a} x^{\frac{1}{2-a}} (1,\infty)$$

$$\text{cdf: } F(x) = \int_0^x \frac{1}{2-a} s^{\frac{1}{2-a}-1} ds = \frac{1}{2-a} x^{\frac{1}{2-a}}$$

$$\text{ITM } \frac{1}{2-a} + \frac{a}{2-a} \int_1^x s^{\frac{1}{2-a}-1} ds$$

$$\textcircled{1} = 1 - \frac{1-a}{2-a} x^{\frac{1}{2-a}} \quad (1, \infty)$$

$$\text{- generate } U \sim \text{unif}(0,1)$$

$$\text{- if } 0 < U < \frac{1}{2-a}, \text{ output } \left(\frac{2-a}{1-a}\right) U^{\frac{2-a}{1-a}}$$

$$\text{else output } \left(\frac{2-a}{1-a}\right) (1-U)^{\frac{2-a}{1-a}}$$

$$\textcircled{2} - U_2 \sim \text{unif}(0,1)$$

$$\text{- } Y = g(x) \cdot U_2 \text{ where } g(x) = g_1(x) \text{ if } x \in (0,1)$$

$$g(x) = g_2(x) \text{ if } x \in (1,\infty)$$

$$g(x) = g_2(x) \text{ if } x \in (1,\infty)$$

$$\text{Composition Method}$$

ex $F(x) = \int_0^\infty x^y e^{-y} dy, 0 \leq x \leq 1$

$$F(x) = \int_0^\infty x^y e^{-y} dy = \int_0^\infty e^{y \ln x - y} dy = \int_0^\infty e^{-y(1 - \ln x)} dy$$

$$\text{- generate } T (-\log(u))$$

$$\text{- given } T=t \quad X \sim F \quad X = (V)^{\frac{1}{t}}$$

$$\text{ex } f_X(x) = \int_0^\infty \frac{9}{(1+xt)^4} t^3 e^{-t^3} dt, x \geq 0$$

$$\text{ex } f_{Y_t}(x) \text{ is not 0}$$

Recomp interpretation

$$I=0 \text{ success}$$

$$X=1 \text{ ; \# of trials}$$

$$\text{while } I=0$$

$$V \sim \text{unif}(0,1)$$

$$\text{if } U < p$$

$$I=1$$

$$\text{else } X=X+1$$

$$\text{return } X$$